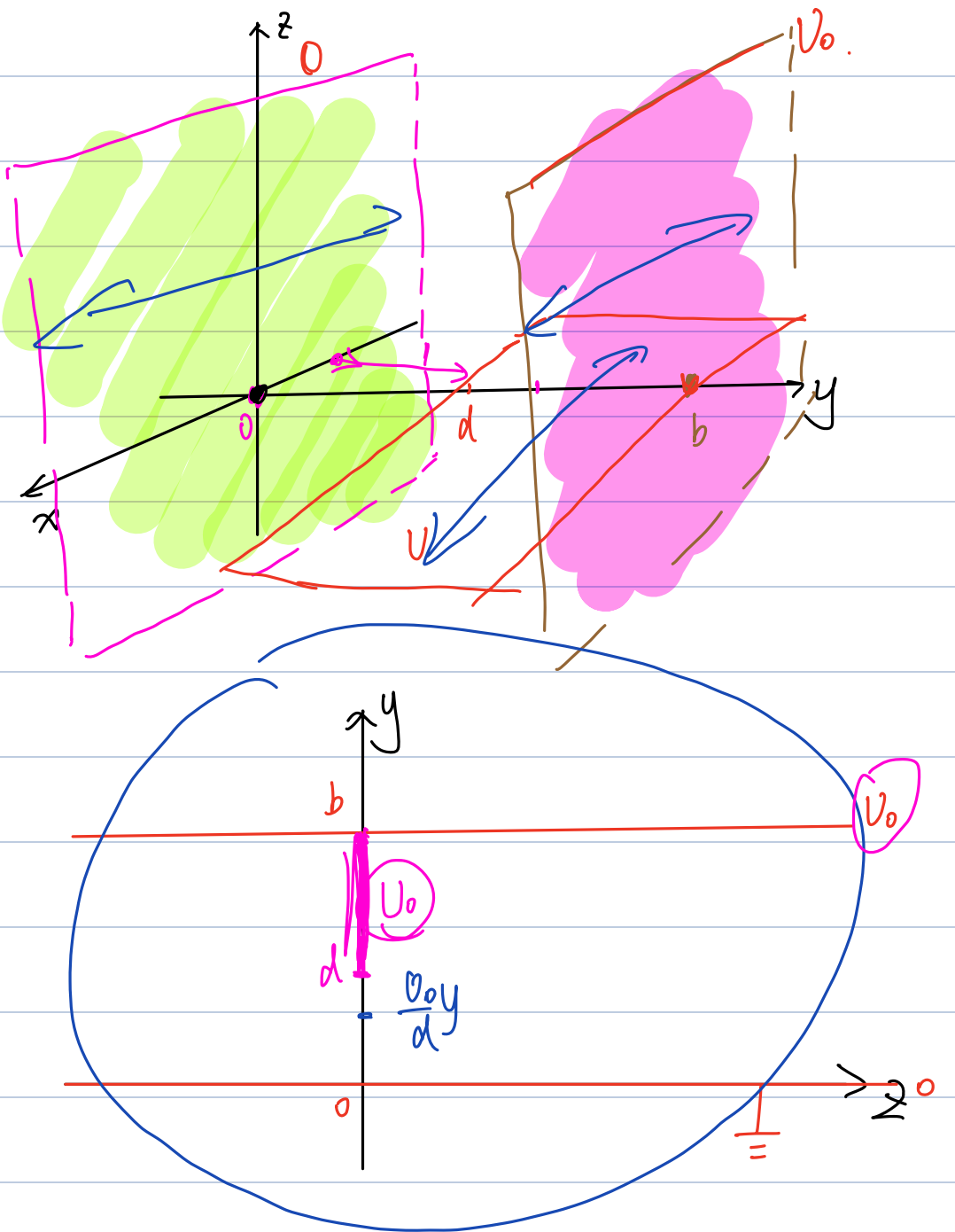
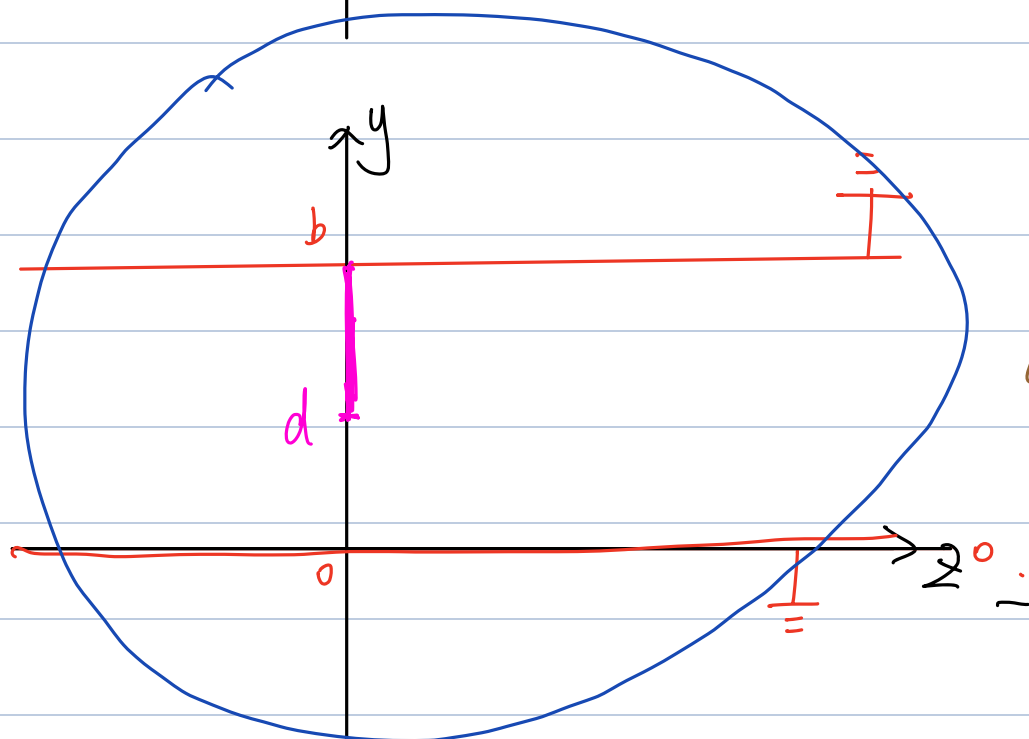
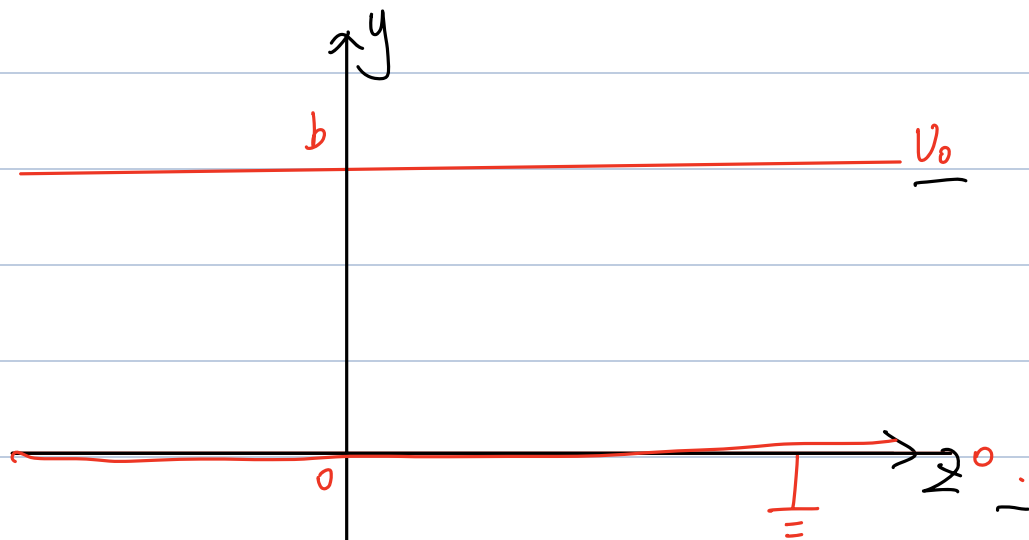


例题 4-7 沿 x, z 方向都极大的两块金属板, 板间距离为 b 。在 $z=0$ 处板间有极薄金属片从 $y=d$ 到 $y=b$ ($-\infty < x < +\infty$)。上板电位为 U_0 , 下板电位为零。在薄平面上, 从 $y=0$ 到 $y=d$ 处 $\varphi = U_0 y/d$, 如图 4-7(a) 所示。试求金属槽内的电位。

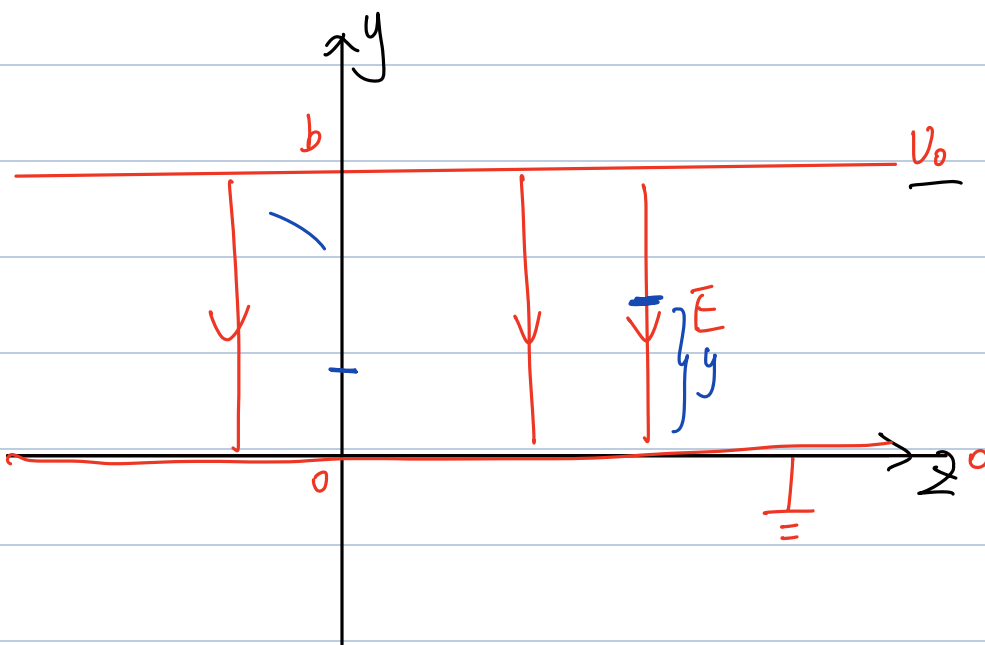


$$\varphi_1 + \varphi_2 = U_0$$



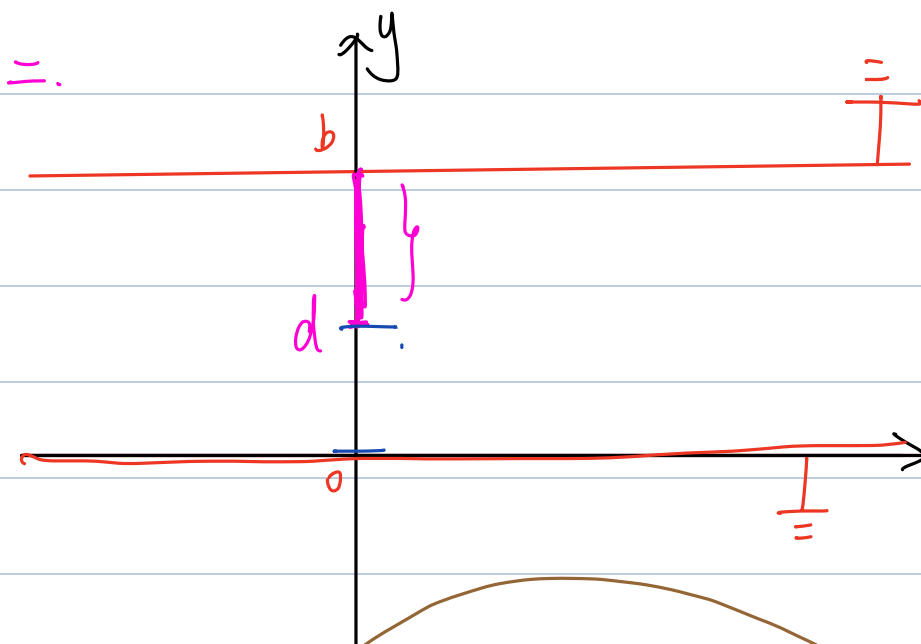
$$\Rightarrow \varphi = \varphi_1 + \varphi_2$$

一. 计算 φ_1



$$E = \frac{U_0}{b}$$

$$\varphi_1 = \frac{U_0}{b} \cdot y$$



$$\nabla^2 \varphi_2 = 0$$

$$\varphi|_{y=b} = 0$$

$$\varphi|_{y=0} = 0$$

$$\varphi|_{z \rightarrow +\infty} = 0$$

$$\varphi|_{z \rightarrow -\infty} = 0$$

$$z=0$$

$$\varphi_2 = \begin{cases} \frac{U_0}{d}y - \frac{U_0}{b}y, & 0 < y < d \\ U_0 - \frac{U_0}{b}y, & d < y < b \end{cases}$$

$$\frac{\partial^2 \varphi}{\partial y^2} + \frac{\partial^2 \varphi}{\partial z^2} = 0$$

$$\varphi(y, z) = Y(y)Z(z)$$

$$ZY'' + YZ'' = 0$$

$$\therefore \frac{Y''}{Y} + \frac{Z''}{Z} = 0$$

$$\frac{Y''}{Y} = -\frac{Z''}{Z} = \lambda$$

$$\begin{cases} Y'' - \lambda Y = 0 \\ Z'' + \lambda Z = 0 \end{cases}$$

先解 $Y'' - \lambda Y = 0$

① $\lambda = 0$ $Y = Ay + B$
 $A = B = 0$

② $\lambda = k^2 > 0$
 $Y = Ae^{kx} + Be^{-kx}$
 $A = B = 0$

③ $\lambda = -k^2 < 0$
 $Y'' + k^2 Y = 0$

$$Y(y) = \underbrace{A \cos ky}_{\downarrow A+0=0} + B \sin ky$$

$$A = 0$$

$$B \sin kb = 0$$

$$kb = n\pi \quad (n = 1, 2, \dots)$$

$$k = \frac{n\pi}{b} \quad \lambda = -\frac{n^2\pi^2}{b^2}$$

$$Y(y) = B \sin\left(\frac{n\pi}{b}y\right)$$

$$Z'' - \frac{n^2\pi^2}{b^2} Z = 0$$

$$Z = C e^{\frac{n\pi}{b}z} + D e^{-\frac{n\pi}{b}z}$$

$$\varphi(y, z) = \sum_{n=1}^{\infty} \left[C_n e^{\frac{n\pi}{b}z} + D_n e^{-\frac{n\pi}{b}z} \right] \cdot \sin\left(\frac{n\pi y}{b}\right)$$

0 ∞

$$z > 0 \quad \varphi(y, z) = \sum_{n=1}^{\infty} D_n e^{-\frac{n\pi}{b}z} \sin\left(\frac{n\pi y}{b}\right)$$

$$\frac{1}{2}z = 0$$

$$\sum_{n=1}^{\infty} D_n \sin\left(\frac{n\pi y}{b}\right) = \begin{cases} \left(\frac{U_0}{d}y - \frac{U_0}{b}y\right) \sin\left(\frac{n\pi y}{b}\right), & 0 < y \leq d \\ \left(U_0 - \frac{U_0}{b}y\right) \sin\left(\frac{n\pi y}{b}\right), & d < y \leq b \end{cases}$$

$$\int_0^b D_n \sin^2\left(\frac{n\pi y}{b}\right) dy = \frac{b}{2} D_n$$

$$= \int_0^d \left(\frac{U_0}{d}y - \frac{U_0}{b}y\right) \sin\left(\frac{n\pi y}{b}\right) dy$$

$$+ \int_d^b \left(U_0 - \frac{U_0}{b}y\right) \sin\left(\frac{n\pi y}{b}\right) dy$$

$$= \left(\frac{U_0}{d} - \frac{U_0}{b}\right) \int_0^d y \sin\left(\frac{n\pi y}{b}\right) dy$$

$$+ U_0 \int_d^b \sin\left(\frac{n\pi y}{b}\right) dy - \frac{U_0}{b} \int_d^b y \sin\left(\frac{n\pi y}{b}\right) dy$$

$$= \frac{U_0}{d} \int_0^d y \sin\left(\frac{n\pi y}{b}\right) dy - \frac{U_0}{b} \int_0^b y \sin\left(\frac{n\pi y}{b}\right) dy$$

$$+ V_0 \int_a^b \sin\left(\frac{n\pi y}{b}\right) dy$$

$$\square = \frac{V_0}{d} \int_0^d y d \cos\left(\frac{n\pi y}{b}\right) \cdot \left(-\frac{b}{n\pi}\right)$$

$$\frac{V_0}{d} \left\{ \left(-\frac{b}{n\pi} y \cos\left(\frac{n\pi y}{b}\right)\right) \Big|_0^d + \frac{b}{n\pi} \int_0^d \cos\left(\frac{n\pi y}{b}\right) d\left(\frac{n\pi y}{b}\right) \frac{b}{n\pi} \right.$$

$$\left. \frac{V_0}{d} \left(-\frac{b}{n\pi} d \cos\left(\frac{n\pi d}{b}\right) + \frac{b^2}{n^2 \pi^2} \int_0^{\frac{n\pi d}{b}} \cos x dx \right) \right.$$

$$\left. \frac{V_0}{d} \left(-\frac{bd}{n\pi} \cos\left(\frac{n\pi d}{b}\right) + \frac{b^2}{n^2 \pi^2} \left[\sin x \Big|_0^{\frac{n\pi d}{b}} \right] \right) \right.$$

$$\left. -\frac{bdV_0}{n\pi} \cos\left(\frac{n\pi d}{b}\right) + \frac{V_0 b^2}{n^2 \pi^2 d} \sin\left(\frac{n\pi}{b} d\right) \right)$$

$$- \frac{V_0}{b} \int_0^b y \sin\left(\frac{n\pi y}{b}\right) dy$$

$$= -\frac{V_0}{b} \int_0^b y d \left[+\cos\left(\frac{n\pi}{b} y\right) \left(-\frac{b}{n\pi}\right) \right]$$

$$= -\frac{V_0}{b} \left[-\frac{b}{n\pi} \cos\left(\frac{n\pi}{b} y\right) y \Big|_0^b + \frac{b}{n\pi} \int_0^b \cos\left(\frac{n\pi}{b} y\right) dy \right]$$

$$= -\frac{V_0}{b} \left[-\frac{b^2}{n\pi} (-1)^n + \frac{b^2}{n\pi^2} \int_0^b \cos\left(\frac{n\pi}{b} y\right) d\left(\frac{n\pi}{b} y\right) \right]$$

$$= \frac{V_0 b}{n\pi} (-1)^n$$

$$V_0 \int_a^b \sin\left(\frac{n\pi y}{b}\right) dy = V_0 \int_a^b \sin\left(\frac{n\pi y}{b}\right) d\left(\frac{n\pi y}{b}\right) \frac{b}{n\pi}$$

$$= \frac{V_0 b}{n\pi} \left[-\cos\left(\frac{n\pi}{b} y\right) \Big|_a^b \right]$$

$$= \frac{V_0 b}{n\pi} \left[-\cos\left(\frac{n\pi}{b} b\right) + \cos\left(\frac{n\pi}{b} a\right) \right]$$

$$= \frac{V_0 b}{n\pi} \left[-(-1)^n + \cos\left(\frac{n\pi d}{b}\right) \right]$$

$$-\frac{bV_0}{n\pi} \cos\frac{n\pi d}{b} + \frac{V_0 b^2}{n^2 \pi^2 d} \sin\left(\frac{n\pi}{b} d\right) + \frac{V_0 b}{n\pi} (-1)^n$$

$$-\frac{V_0 b}{n\pi} (-1)^n + \frac{V_0 b}{n\pi} \cos\left(\frac{n\pi d}{b}\right)$$

$$\frac{b}{2} D_n = \frac{V_0 b^2}{n^2 \pi^2 d} \sin\left(\frac{n\pi}{b} d\right)$$

$$D_n =$$

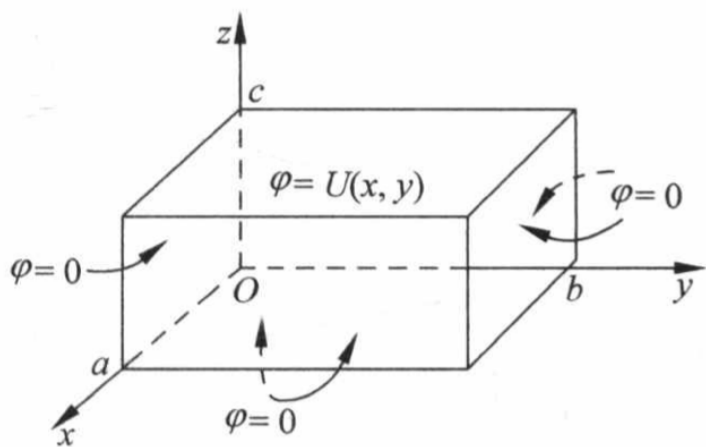
$$z > 0$$

$$\varphi(y, z) = \frac{V_0}{b} y + \sum_{n=1}^{\infty} \frac{2V_0 b}{n^2 \pi^2 d} \sin\left(\frac{n\pi}{b} d\right) e^{-\frac{n\pi}{b} z} \sin\left(\frac{n\pi y}{b}\right)$$

$$2co \quad \boxed{\varphi_1 =}$$

$$\varphi(y, z) = \frac{V_0}{b} y + \sum_{n=1}^{\infty} \frac{2V_0 b}{n^2 \pi^2 d} \sin\left(\frac{n\pi}{b} d\right) e^{\frac{n\pi z}{b}} \sin\left(\frac{n\pi y}{b}\right)$$

例题 4-8 一个长方形导体盒，各边尺寸分别是 a 、 b 、 c ，其四周和底部为零电位，顶部与其他周界相互绝缘，电位函数是 $U(x, y)$ ，如图 4-8 所示。试求导体盒内部的电位函数。



$$\nabla^2 \varphi = 0$$

$$\frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2} + \frac{\partial^2 \varphi}{\partial z^2} = 0$$

$$\varphi|_{y=0} = 0$$

$$\varphi|_{x=0} = 0$$

$$\varphi|_{y=b} = 0$$

$$\varphi|_{z=0} = 0$$

$$\varphi|_{x=a} = 0$$

$$\varphi|_{z=c} = U(x, y)$$

$$\varphi(x, y, z) = X(x) Y(y) Z(z)$$

$$YZX'' + XZY'' + XYZ'' = 0$$

$$\frac{X''}{X} + \frac{Y''}{Y} + \frac{Z''}{Z} = 0$$

$$\alpha + \beta + \gamma = 0$$

$$\begin{cases} X'' - \alpha X = 0 \\ Y'' - \beta Y = 0 \\ Z'' - \gamma Z = 0 \end{cases}$$

$$\text{先解 } Y'' - \beta Y = 0$$

$$\beta = -k_y^2 < 0$$

$$Y'' + k_y^2 Y = 0$$

$$\therefore Y(y) = A \cos(k_y y) + B \sin(k_y y)$$

$$A + B \cdot 0 = 0$$

$$\Rightarrow A = 0$$

$$B \sin(k_y b) = 0$$

$$k_y b = n\pi$$

$$k_y = \frac{n\pi}{b}$$

$$Y(y) = B \sin\left(\frac{n\pi}{b} y\right)$$

$$\beta = -\frac{n^2 \pi^2}{b^2}$$

$$\text{再解 } X'' - \alpha X = 0$$

$$\alpha > 0 \quad \alpha = 0$$

$$\alpha = -k_x^2 < 0$$

$$X'' + k_x^2 X = 0 \quad D \sin(k_x a) = 0$$

$$X(x) = C \cos(k_x x) + D \sin(k_x x)$$

$$C + 0 = 0$$

$$C = 0$$

$$k_x a = n\pi$$

$$k_x = \frac{n\pi}{a}$$

$$X(x) = D \sin\left(\frac{n\pi}{a} x\right) \quad (n_1 = 1, 2, \dots)$$

$$\alpha = -\frac{n^2 \pi^2}{a^2}$$

$$r = -\alpha - \beta = \left(\frac{\hbar^2 \lambda^2}{a^2} + \frac{\hbar^2 \lambda^2}{b^2} \right)$$

$$Z'' - rZ = 0$$

$$Z(z) = E e^{\sqrt{r}z} + F e^{-\sqrt{r}z}$$

$$\psi(x, y, z) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} (E e^{\sqrt{r}z} + F e^{-\sqrt{r}z}) \sin\left(\frac{m\pi}{a}x\right) \cdot \sin\left(\frac{n\pi}{b}y\right)$$

$$\sum_{n=1}^{\infty} \sum_{m=1}^{\infty} (E + F) \sin(x) \sin(y) = 0$$

$$E_n + F_n = 0$$

$$\int_0^a \int_0^b \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} (E e^{\sqrt{r}z} + F e^{-\sqrt{r}z}) \sin^2\left(\frac{m\pi}{a}x\right) \sin^2\left(\frac{n\pi}{b}y\right) dy dx = \iint V(x, y) \sin\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) dy dx$$

$$(E e^{\sqrt{r}z} + F e^{-\sqrt{r}z}) \left(\frac{a}{z} \cdot \frac{b}{z} \right) = \int_0^a \int_0^b V(x, y) \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) dx dy$$

$$E_n e^{\sqrt{r}z} + F_n e^{-\sqrt{r}z} = \frac{4}{ab} \int_0^a \int_0^b V(x, y) \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) dx dy$$

G